Daehyun Jang

CV Ver:
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Seoul, Republic of Korea jadh0309@snu.ac.kr

ABOUT ME

I am a Ph.D. student in Mathematical Sciences at Seoul National University, advised by Prof. Jung Hee Cheon. My research focuses on Cryptography, particularly Homomorphic Encryption.

EDUCATION

Integrated M.S./Ph.D. Program in Mathematical Sciences Sep. 2024 – Present
Seoul National University, Seoul, Republic of Korea

B.S. in Mathematical Sciences Mar. 2019 – Aug. 2024
Seoul National University, Seoul, Republic of Korea

- *GPA*: 3.86/4.3 (Overall)
  - *Major*: Mathematical Sciences (3.99/4.3)
  - *Minor*: Philosophy (3.80/4.3)
- *Military Service* at Republic of Korea Army Dec. 2020 – Jun. 2022

PUBLICATIONS

Published

- Jung Hee Cheon, **Daehyun Jang**, Jaehee Kang, Hanee Rhee, "Private Embedding Lookup with Encrypted Compact Queries under Fully Homomorphic Encryption," *Proceedings on Privacy Enhancing Technologies (PoPETs) 2027, Issue 1*.
  Accepted for publication; author order is alphabetical by surname.
- Jung Hee Cheon, **Daehyun Jang**, "Cryptanalysis on Lightweight Verifiable Homomorphic Encryption," *Asiacrypt 2025*.
  Author order is alphabetical by surname.

Preprint

- **Daehyun Jang**, Junho Lee, "Arithmetic for Large-Characteristic Finite Fields in CKKS," Cryptology ePrint Archive, Paper 2026/1958.
  Author order is alphabetical by surname.
- Jung Hee Cheon, **Daehyun Jang**, Jaehee Kang, "SWIFT: Shallow and SIMD-Aware CKKS Functional Bootstrapping for Low-Latency"
  Author order is alphabetical by surname.

TALKS

- Korean Mathematical Society (KMS) 2024 Fall Meeting
  "Cryptanalysis on Lightweight Verifiable Homomorphic Encryption"
- Asiacrypt 2025
  "Cryptanalysis on Lightweight Verifiable Homomorphic Encryption"
- International Conference for the 80th Anniversary of the Korean Mathematical Society
  Jun. 23, 2026
  "SWIFT: Shallow and SIMD-Aware CKKS Functional Bootstrapping for Low-Latency"

- PACOH Workshop: Homomorphic Encryption 2026
  Jun. 29, 2026, Sokcho, Republic of Korea
  "SWIFT: Shallow and SIMD-Aware CKKS Functional Bootstrapping for Low-Latency"

POSTERS

- FHE.org Conference 2026, "Efficient Private Embedding Lookup via Independent Vector Evaluation"

RESEARCH INTERNSHIP

Graduate Research Fellow

Korea Institute for Advanced Study (KIAS), Seoul, Republic of Korea

- Homomorphic Encryption — Jan. 2025 – Present
  - Theoretical study on the CKKS scheme
  - Advised by Prof. Jeong Han Kim

Undergraduate Research Internship

Seoul National University, Seoul, Republic of Korea

- Homomorphic Encryption — Oct. 2023 – Aug. 2024
  - Study on the CKKS scheme and its recent advances
  - Advised by Prof. Jung Hee Cheon
- Homogeneous Dynamics — Jun. 2023 – Aug. 2023
  - Study on the Quantitative version of the Oppenheim conjecture
  - Advised by Prof. Seonhee Lim

TEACHING EXPERIENCE

Teaching Assistant

- Cryptography — Fall 2026
- Abstract Algebra I — Spring 2026
- Engineering Mathematics I — Spring 2026
- Mathematical Exercises II — Fall 2024, Fall 2025
- Advanced Mathematical Exercises I — Spring 2025

Undergraduate Tutor

- Mathematics I — Spring 2020, Spring 2024
- Mathematics II — Fall 2020
- Mathematics for Life Sciences I — Spring 2023
- Calculus II — Fall 2023

HONORS & AWARDS

- 3rd Place, Korea Crypto Contest — Oct. 2026
- 3rd Place, Korea Crypto Contest — Oct. 2025
- Outstanding Teaching Assistant Award, Seoul National University — Dec. 2024
- Outstanding Tutor Award, Seoul National University — Dec. 2023
- Outstanding Student Award, Seoul National University — Dec. 2020